ABSTRACT

Efficient traffic management plays a critical role in ensuring a timely response for emergency vehicles such as ambulances, fire trucks, and police cars. Traditional traffic systems often fail to provide the necessary prioritization, leading to delays that can cost lives. This report presents the development of an AI-based emergency vehicle detection and traffic light control system designed to address this challenge. Leveraging real-time video feed analysis using deep learning techniques, the system accurately detects emergency vehicles approaching intersections. Once identified, the system dynamically adjusts traffic signals to provide a clear path, minimizing delays and improving response times. The model is trained on diverse datasets to ensure robustness in various lighting and weather conditions. Simulation results demonstrate a significant reduction in waiting time for emergency vehicles, proving the system's effectiveness and potential for real-world deployment. This AI-powered approach not only enhances emergency response efficiency but also contributes to smarter, safer, and more adaptive traffic management.

PROBLEM STATEMENTS

• The existing method used to control traffic is the Traffic light System. A numerical value is loaded in the timer for each phase. Depending on changes in the timer, the light is set to ON and OFF states on different lanes. The problem with this approach is that it is not good to have a green light on an empty lane.

• Another method used to control traffic is by using sensors. Sensors will get the traffic information about a particular lane, and accordingly, traffic lights might be set. However, the problem is that the traffic information provided by sensors is limited.

• The main aim is to implement a traffic management system that allocates the time to each lane based on the density and also to provide maximum priority to the lane where the ambulance arrives.

OBJECTIVES

To provide a clear way to the ambulance whenever it enters into the range of camera and to control the signals by measuring the density of traffic thereby avoiding the wastage of time and saving the lives of human being.

1. Deployment of an adaptive visual sensing framework that utilizes real-time image processing techniques to quantify vehicular density across multiple lanes, enabling responsive traffic regulation.
2. Development of a dynamic traffic signal control mechanism that modifies signal timing based on per-lane traffic density, aiming to reduce wait times and optimize vehicular throughput.
3. Implementation of an emergency vehicle recognition and prioritization feature that accurately identifies such vehicles and adjusts signal phases accordingly to allow swift and uninterrupted passage.
4. Design of a lightweight, browser-based monitoring interface to provide operators with real-time visualization and system interaction capabilities.

LITERATURE SURVEY

1.Enhancing Urban Traffic Management with Real-Time Computer Vision and AI: A Smart Traffic Signal System

This paper introduces an AI-driven smart traffic signal system utilizing YOLO-based real-time object detection and OpenCV to manage urban traffic dynamically. The system effectively performs multi-lane vehicle counting, prioritizes emergency vehicles, and automates lane control using motorized barriers. A major contribution is the integration of a lane optimization algorithm, which minimizes waiting time across all lanes based on real-time data. Simulation and real-world testing validate the model's capability to reduce congestion and improve safety. The model's scalability is also addressed, and challenges related to deployment are discussed, such as environmental variability and data reliability. This study builds upon previous research by improving real-time responsiveness and offering automated, scalable solutions for smart urban mobility. An Automated Vehicle Counting System Based on Blob Analysis for Traffic Surveillance [1] by G. Salvi.

2. AI-Based Traffic Flow Prediction for Smart Urban Mobility

This study presents an integrated approach using the latest YOLOv9 model to address Indian urban traffic issues, which are often characterized by a wide variety of vehicle types and dense conditions. It proposes a novel method of traffic density estimation based not just on vehicle count but on the spatial length of traffic using bounding boxes defined by road features. In addition to signal timing, the system features accident detection, emergency vehicle recognition, and peak hour prediction using ARIMA models. Comparative analysis showed a significant increase in vehicle detection accuracy (from 28.57% using YOLOv8 to 89.29% with the proposed method). By combining multiple previously isolated features into a unified model, this paper emphasizes real-world adaptability, cost efficiency, and predictive capabilities for Indian traffic systems.

3. Advanced Traffic Control System Using AI

This paper proposes a rule-based Expert System for traffic signal control, called the Traffic Lights Expert System (TLES). It uses evidential reasoning and blob detection techniques for vehicle tracking and categorization. The system is enhanced with ZigBee communication to manage pre-emption for emergency vehicles, making it responsive to real-time route clearance. A key strength is the low-cost hardware setup (Arduino, Raspberry Pi, XBee) that supports image processing through OpenCV.

4. Infrared and IoT-Based Dynamic Traffic Control

The study by Rani et al. (2017) proposed a Dynamic Traffic Management System using Infrared (IR) and IoT that automates signal control based on traffic density detected through IR sensors. Data from sensors are transmitted to a Raspberry Pi controller, which dynamically alters signal durations—allocating more green light time to congested roads. The system includes a mobile interface to alert users of signal status and integrates cloud storage for historical traffic data. Although efficient for low-cost deployment, limitations exist in sensor accuracy and real-time scalability, particularly in high-traffic or multi-lane environments.

5. Computer Vision and AI-Powered Smart Signal System

Salman et al. (2024) introduced a Smart Traffic Signal System using Computer Vision and AI to address urban traffic inefficiencies more robustly. The system employs a pre-trained YOLO object detection model and OpenCV to perform real-time vehicle and lane detection. Advanced features include:

Multi-lane vehicle counting. Emergency vehicle prioritization.

Optimization algorithms for dynamic green light allocation.

Motorized barriers to reduce manual intervention.

Simulation and field tests demonstrated significant improvements in traffic flow and adaptability. The system also showcased strong potential for scalability in larger urban environments, although challenges remain in integrating additional sensory inputs (e.g., LIDAR) and ensuring consistent performance under varied environmental conditions.

6. Hybrid Model Using Infrared Sensors and Acoustic Detection

Saxena and Jebakumari (2023) proposed a Smart Traffic System combining IR sensors and Bell Detection mechanisms to enhance adaptive control capabilities. The system integrates:

IR sensors for real-time vehicle presence detection.

A microcontroller with machine learning capabilities for decision-making.

7. IoT for Traffic Law Enforcement and Smart Regulation

Mehta and Singh (2024) present an advanced IoT-enabled vehicle control system tailored for traffic law enforcement and smart urban mobility. Their system integrates real-time data collection from cameras, RFID readers, and speed sensors, processed by cloud-based machine learning algorithms to detect violations like speeding and red-light offenses. The system demonstrated a 92.3% detection rate with 30% fewer violations reported during its three-month trial. Additionally, it improved emergency vehicle response time by 18% and traffic flow efficiency by 15%.

The architecture employs:

CNNs for visual violation detection.

RNNs for traffic forecasting. Random Forest models for vehicle classification and severity analysis.

Key challenges included internet dependency, integration with legacy traffic systems, and public concerns about data privacy. Nonetheless, the system showcased how real-time IoT and AI technologies can shift enforcement from reactive to proactive.

8. Smart Self-Powered Street Lighting with Road Safety Functions

Islam et al. (2021) explore a multi-functional IoT-based street lighting system that integrates solar energy, automated lighting control, accident detection, and emergency alert mechanisms. Their design uses renewable energy (solar panels), LDRs, ultrasonic sensors, and microcontrollers (Node MCU) to create an intelligent and energy-efficient urban lighting system.

Noteworthy features include:

Automated intensity control of street lights based on ambient lighting and vehicle presence.

Accident detection via ultrasonic sensors with immediate notification to authorities.

Emergency push-button with integrated face recognition for real-time distress alerts.

Nameplate detection of vehicles using image processing for accident identification.

This system emphasizes sustainable energy use and public safety, offering a cost-effective solution for under resourced areas. Although still in a prototype stage, it addresses gaps in existing systems by combining energy efficiency with real-time safety interventions.

METHODOLOGY

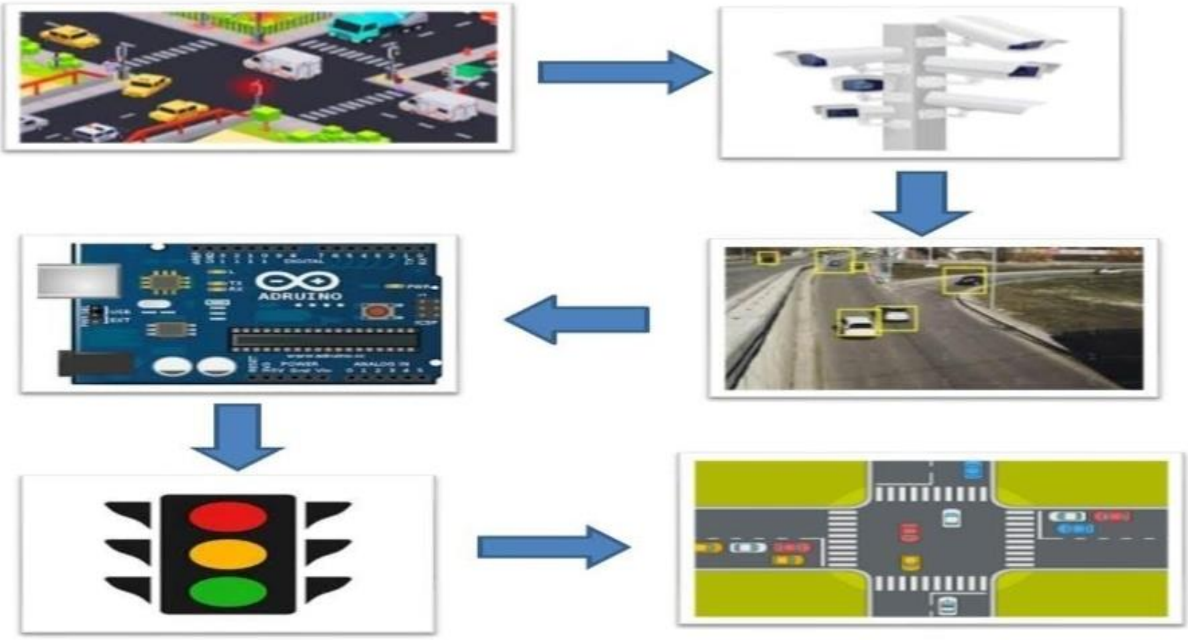


Figure 1: Block Diagram of Proposed Methodology

Gathering requirements is the main attraction of the Analysis Phase. The process of gathering requirements is usually more than simply asking the users what they need and writing their answers down. Depending on the complexity of the application, the process for gathering requirements has a clearly defined process of its own. This process consists of a group of repeatable processes that utilize certain techniques to capture, document, communicate, and manage requirements. Systems design is the process of defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements. Systems design could be seen as the application of systems theory to product development. There is some overlap with the disciplines of systems analysis, systems architecture, and systems engineering.

If the broader topic of product development "blends the perspective of marketing, design, and manufacturing into a single approach to product development," then design is the act of taking the marketing information and creating the design of the product to be manufactured. Systems design is, therefore, the process of defining and developing systems to satisfy the specified requirements of the user.

Until the 1990s, systems design had a crucial and respected role in the data processing industry. In the 1990s, the standardization of hardware and software resulted in the ability to build modular systems. The increasing importance of software running on generic platforms has enhanced the discipline of software engineering.

Object-oriented analysis and design methods are becoming the most widely used methods for computer systems design. The Open CV has become the standard software in analysis and design. It is widely used for modelling software systems and is increasingly used for high designing non-software systems and organizations. System design is one of the most important phases of the software development process. The purpose of the design is to plan the solution to a problem specified by the requirement documentation. In other words, the first step in the solution to the problem is the design of the project.

The design of the system is perhaps the most critical factor affecting the quality of the software. The objective of the design phase is to produce the overall design of the software. It aims to figure out the modules that should be in the system to fulfill all the system requirements efficiently.

The design will contain the specification of all these modules, their interaction with other modules, and the desired output from each module. The output of the design process is a description of the software architecture. The design phase is followed by two sub-phases:

- High-Level Design.
- Low-Level Design

HARDWARE AND SOFTWARE REQUIREMENTS

HARDWARE SPECIFICATIONS

The most common type of requirements defined by any operating system or software application is the physical computer resources, also known as hardware. The hardware requirement list is:

- PC with open CV Python
- Web Camera
- Microcontroller-Raspberry Pi
- USB to UART
- LCD
- LED
- Buzzer

SOFTWARE SPECIFICATIONS

The software requirements are descriptions of features and functionalities of the system. The software requirement is a field within software engineering that deals with establishing the needs of stakeholders that are to be solved by software. Requirements convey the expectations of users from the software product. The software requirement list is:

- Operating System: Windows 7 and above.
- Software Module: OpenCV Image Processing.
- Interfacing: The interfacing between the hardware.
- Computer: A general-purpose PC serves as a central processing unit for various tasks.

HARDWARE DESCRIPTION

• Raspberry Pi

The Raspberry Pi is designed to be connected to the Internet. Its ability to communicate on the Internet is one of its key features and opens up all sorts of possible uses, including home automation, web serving, network monitoring, and so on. The connection can be wired through an Ethernet cable, or the Pi can use a USB Wi Fi module to provide a network connection. This is very useful in situations where the Raspberry Pi itself is accessible and does not have a keyboard, mouse, or monitor attached to it. Specifications:

CPU: 4x ARM Cortex-A53, 1.2 GHz.

GPU: Broadcom Video Core IV.

RAM: 1GB LPDDR2(900MHZ).

Networking: 10/100 Ethernet,2.4 GHz 802.11n wireless.

Bluetooth: Bluetooth 4.1 Classic, Bluetooth Low Energy.

Storage: Micro SD.

GPIO: 40-pin header, populated.

Ports: HDMI, 3.4mm analogue audio-video jack, 4x USB 2.0 Ethernet, Camera Serial Interface (CSI). Display Serial Interface (DSI).

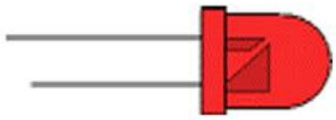
• Bus strips

To provide power to the electronic components. A bus strip usually contains two rows: one for ground and one for a supply voltage. However, some breadboards only provide a single-row powered distribution bus strip on the long side. Typically, the row intended for a supply voltage is marked in red, while the row for ground is marked in blue or black. Some manufacturers connect all terminals in a column. Others just connect groups of, for example, 25 consecutive terminals in a column. The latter design provides a circuit designer with some more control over crosstalk on the power supply bus. Often, the groups in a bus strip are indicated by gaps in the color marking.

Bus strips typically run down one or both sides of a terminal strip or between terminal strips. On large breadboards, additional bus strips can often be found on the top and bottom of terminal strips.

• LED

LEDs are semiconductor devices. Like transistors and other diodes, LEDs are made out of silicon. What makes an LED give off light are the small amounts of chemical impurities that are added to the silicon, such as gallium, arsenide, indium, and nitride. When the current passes through the LED, it emits photons as a byproduct. Normal light bulbs produce light by heating a metal filament until it is white hot. LEDs produce photons directly and not via heat, they are far more efficient than incandescent bulbs.



Not long ago, LEDs were only bright enough to be used as indicators on dashboards or electronic equipment. But recent advances have made LEDs bright enough to rival traditional lighting technologies. Modern LEDs can replace incandescent bulbs in almost any application.

• Liquid Crystal Display

A liquid crystal display (LCD) is a thin, flat panel used for electronically displaying information such as text, images, and moving pictures. Its uses include monitors for computers, televisions, instrument panels, and other devices ranging from aircraft cockpit displays to everyday consumer devices such as video players, gaming devices, clocks, watches, calculators, and telephones.

SOFTWARE DESCRIPTION

Open CV

OpenCV is an open-source C++ library for image processing and computer vision originally developed by Intel and now supported by Willow Garage. It is free for both commercial and non-commercial use. Therefore, it is not mandatory for your OpenCV communication to open for free; it is a library of many inbuilt functions mainly aimed at real-time image processing. Now, it has several hundred image processing and computer vision algorithms, which make developing advanced computer vision applications easy and efficient. If you are having any trouble with installing OpenCV or configuring your Visual Studio IDE for OpenCV, please refer to Installing and Configuring with Visual Studio.

FLOW CHART

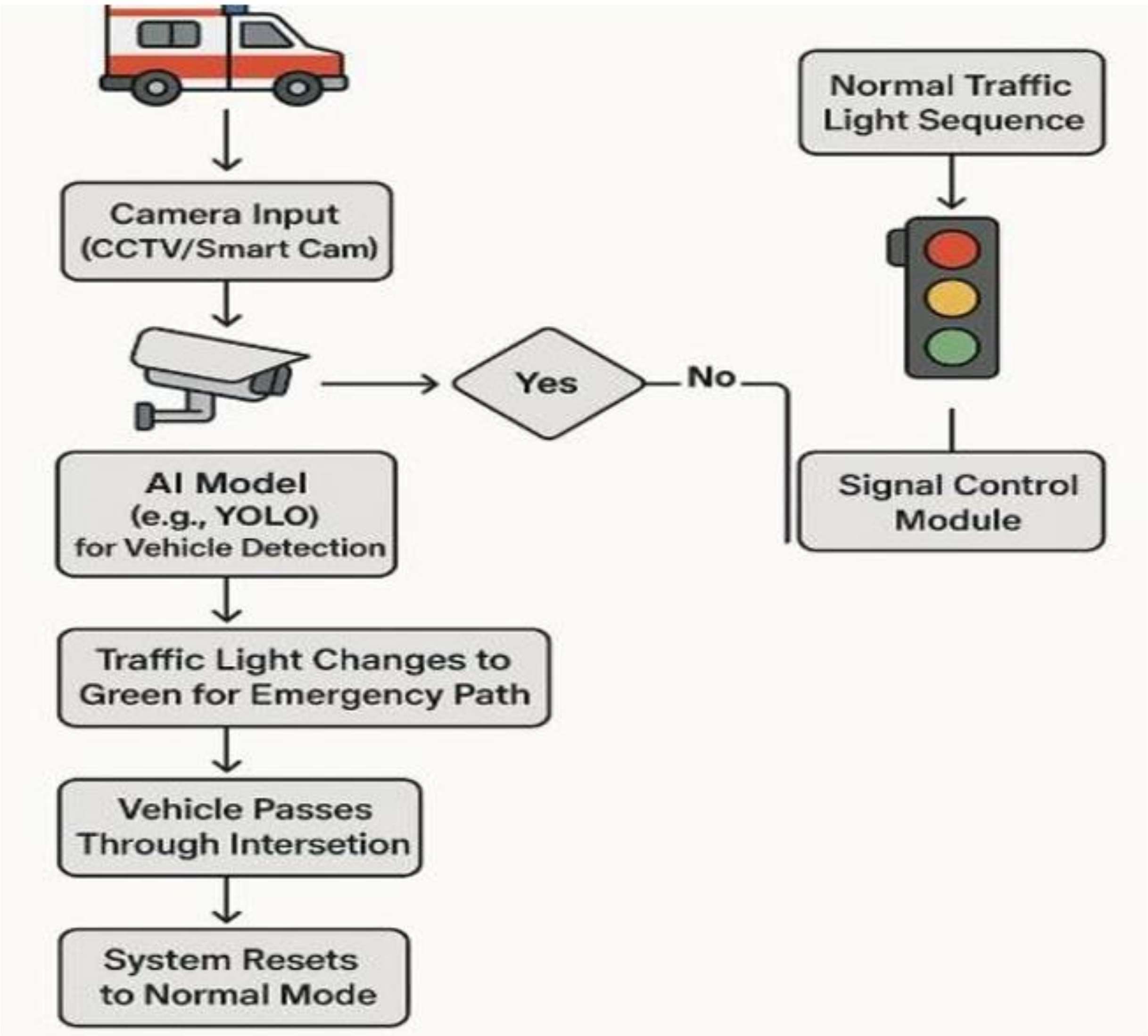


FIGURE 2 FLOW CHART OF THE TRAFFIC MANAGEMENT SYSTEM

Analysis is the process of breaking a complex topic or substance into smaller parts to gain a better understanding of it. Analysts in the field of engineering look at requirements, structures, mechanisms, and systems dimensions. Analysis is an exploratory activity. The Analysis Phase is where the project lifecycle begins.

The Analysis Phase is where you break down the deliverables in the high-level Project Charter into the more detailed business requirements.

RESULT AND DISCUSSION

1. Analysis of Vehicle Identification Performance

The vehicle identification component was trained using a comprehensive dataset containing images with varied numbers and types of vehicles. Evaluation results indicated that the model achieved an accuracy between 80% and 85% in detecting vehicles across different conditions. Minor inaccuracies were observed in densely populated scenes. Incorporating real-world traffic video data is expected to improve model generalization and detection fidelity**.**

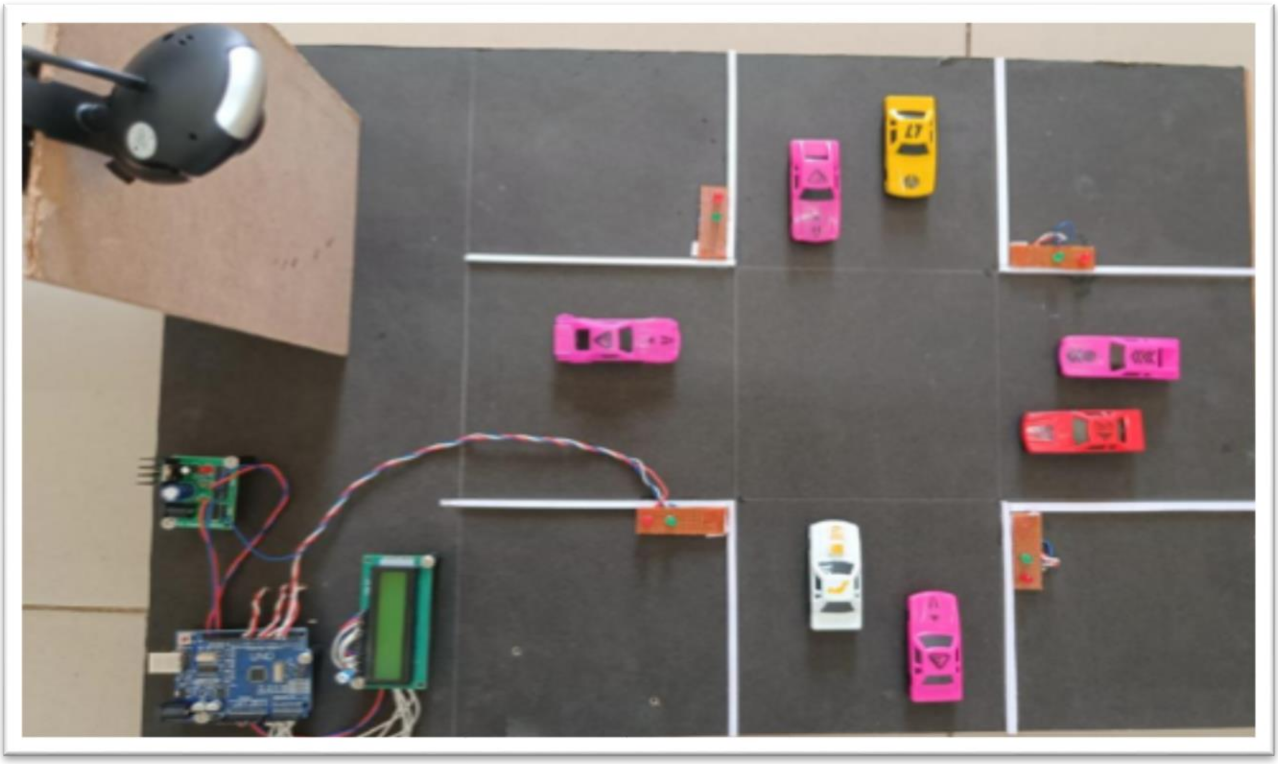


FIGURE 3 PROPOSED SYSTEM MODEL

2. Assessment of Priority Allocation Logic

The module responsible for priority determination was tested using images that included both standard vehicles and high-priority units such as ambulances. The system effectively distinguished emergency vehicles and assigned appropriate priorities, reaching an accuracy rate of 80% to 90%. These results affirm the module's capability to make context-aware priority decisions. Future work will involve enhancing the dataset with real-time traffic scenarios to further boost accuracy.

FIGURE 4 VEHICLE DETECTION AND PRIORITY ASSIGNMENT MODEL

3. Evaluation of Adaptive Signal Management System

Due to practical deployment limitations, the traffic control logic was validated within a simulation environment. The system successfully identified lanes with greater congestion and those containing emergency vehicles, assigning them green signals accordingly. The module achieved an operational accuracy between 70% and 80%, indicating strong potential for real-time traffic regulation.

APPLICATIONS

• Urban Traffic Management: Helps reduce congestion in cities by optimizing signal timings based on real-time traffic data.

• Emergency Vehicle Priority: Ensures faster and safer passage for ambulances, fire trucks, and police vehicles by providing a green corridor.

• Accident Reduction at Intersections: Minimizes the risk of collisions, especially involving emergency vehicles, by intelligently controlling traffic flow.

• Smart City Infrastructure: Acts as a core component of smart transportation systems, integrating with other urban technologies for holistic traffic solutions.

• Environmental Benefits: Reduces vehicle idling time at signals, leading to lower fuel consumption and decreased air pollution.

• Event and Rush Hour Management: Effectively handles traffic surges during events, peak hours, or road maintenance through adaptive signal control.

ADVANTAGES

• Efficient Object Detection with YOLOv5:
Uses the YOLOv5 algorithm, known for its real-time speed and high accuracy, to detect vehicles and emergency vehicles quickly and accurately.

• Real-time Dynamic Signal Control: Based on live traffic density and vehicle type, the system adjusts traffic light timings instantly, reducing congestion and travel time.

• Emergency Vehicle Priority: The system detects ambulances or fire trucks using YOLOv5 and immediately assigns them the highest signal priority.

• Low-Cost Hardware with Arduino: Arduino is used to control traffic lights, making the system affordable, modular, and easy to implement at intersections.

• Uses Existing Infrastructure: No need for expensive new sensors—just standard CCTV cameras and a basic hardware setup

• Environmentally Friendly: Reduces unnecessary idling and wait time , leading to lower fuel consumption and pollution.

CONCLUSION

With this project, the problem of traffic can be easily sorted out: the timing of each signal can be automatically adjusted according to the density of traffic, which is real-time operation. It will also clear the path for the ambulance and fire brigade in emergency cases, and it will also help the public in taking decisions for reaching their destination in time using auto routing method. It shows that it can reduce the traffic congestion and avoid wasting time by a green light on an empty road. It is also more consistent in detecting vehicle presence because it uses actual traffic images. It visualizes the reality, so it functions much better than those systems that rely on the detection of the vehicle's metal content. Overall, the system is good, but it still needs improvement to achieve one hundred percent accuracy. The proposed AI-based traffic light system demonstrates a promising approach to mitigating urban traffic congestion by dynamically adjusting signal priorities based on real-time traffic density and the presence of emergency vehicles. Leveraging existing infrastructure such as CCTV cameras and pre-trained machine learning models, the system optimizes signal timings to reduce vehicle waiting times and improve traffic flow efficiency. Simulation results indicate notable improvements in traffic management, making the solution.

FUTURE SCOPE

In future work, the focus will be on enhancing the system's detection accuracy by training the algorithms with real-time traffic footage, including low-light and adverse weather scenarios. Furthermore, integration with vehicle-to-infrastructure (V2I) communication protocols will enable faster identification of emergency vehicles. Expanding the system to support inter-junction communication will also be considered to allow synchronized traffic control across multiple intersections. Additionally, deploying the system on edge computing platforms will improve latency and decision-making speed, making it more suitable for large-scale implementation in smart cities.

- Smart City Integration: The system can be integrated with existing smart city infrastructure to enable centralized monitoring, real time data sharing, and coordinated urban mobility solutions.
- Predictive Traffic Management: By implementing machine learning algorithms, the system can analyze historical and real-time traffic data to predict congestion and optimize signal timings in advance.
- Mobile Application Development: A dedicated mobile app can facilitate direct communication between emergency vehicles and the traffic control system, ensuring a quicker and more reliable green corridor.
- Multi-Junction Coordination: Future versions of the system can manage multiple intersections simultaneously, ensuring a smooth and continuous flow of traffic and emergency vehicles across larger urban networks.
- Integration with IoT Devices: Incorporating Internet of Things (IoT) devices like smart sensors and connected vehicles can enhance data accuracy and system responsiveness, making traffic control even more efficient and adaptive.

REFRENCES

[1] Brihadeeswaran N, Ms. Prathipa S, Krishna Vamsi K, “AI-powered Traffic Lights”, ICCEBS, 2023.

[2] P. Sai Srujan Reddy, B.U Naveen Raj, Nikhil Tom Jose, Meena Belwal “Advanced Adaptive Traffic Management System for Urban Environments”, International Conference IDCIoT, 2025.

[3] M M Salman, Md. Mahbubullah Akil, Shanjida Islam, Md Nurul Absar Siddiky, Muhammad Enayetur Rahman, and Md. Shamim Ahsa, “Enhancing Urban Traffic Management with Real-Time Computer Vision and AI: A Smart Traffic Signal System”, IEEE International Conference on SPICSCON, 2024.

[4] Ashwini Bhosale, Arshiya Shaikh, et al., “AI-Based Traffic Flow Prediction for Smart Urban Mobility”, ICAIQSA, 2024.

[5] Ashendra Kumar Saxena, Adlin Jebakumari S, “A Smart Model to Manage Traffic using Infrared Sensors and Bell Detection System: A Computer Controlled Traffic System”, 2023.

[6] Qi Jin. “Automatic Control of Traffic Lights at Multiple Intersections Based on AI and ABST Light”, IEEE, 2024.

[7] Rashmi Rani Samanta Ray, Abdul Azeez, Nayana Hegde S “Advanced Traffic Control System using AI”, IITCEE, 2025.

[8] Shiva Mehta, Manpreet Singh, “Revolutionizing Traffic Law Enforcement: IoT-Enabled Vehicle Control Systems for Smart Cities”, IEEE, 2024.

[9] Md. Hasibul Islam et al. “IoT-Based Smart Self-Power Generating Street Light and Road Safety System Design”, IEEE, 2021.

[10] Jain, Naman, Rachit Parwanda, and Anamika Chauhan. "Real-Time Smart Traffic Control and Simulation”, Congestion Management, IEEE, 2023.

[11] K F Chu, A Y S Lam, and V O K Li, “Traffic signal control using end-to-end off-policy deep reinforcement learning”, IEEE, 2022.

[12] V Upadhyay and N. Sivakumar, "Traffic Monitoring System using YOLOv3 Model", 2023 7th International Conference on Intelligent Computing and Control Systems (ICICCS), 2023.

[13] M. P. Mohandass, K. I, M. R, and V. R, "IoT-Based Traffic Management System for Emergency Vehicles", (ICACCS) 2023.

[14] D. T. Mane, S. Sangve, S. Kandhare, S. Mohole, S. Sonar, and S. Tupare, "Real-Time Vehicle Accident Recognition from Traffic Video Surveillance using YOLOV8 and OpenCV", International Journal on Recent and Innovation Trends in Computing and Communication, 2023